Nuzhat Nuari Khan Rivu

755 West Benton St
Iowa City,IA 52246

+19254366145
nuzhatnuarikhan-rivu@uiowa.edu
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Education

- **PhD in Applied Mathematical and Computational Sciences** *Aug, 2024-Present*
  Department of Mathematics, University of Iowa
- **Master of Science in Mathematics** *Apr, 2023-May, 2024*
  Department of Mathematics, University of Dhaka CGPA: 3.81/4.00
- **Bachelor of Science in Mathematics** *Jan, 2018-Jan, 2023*
  Department of Mathematics, University of Dhaka CGPA: 3.60/4.00
- **Higher Secondary School Certificate (HSC)** *2015-2017*
  Viqarunnisa Noon School and College GPA: 5.00/5.00
- **Secondary School Certificate (SSC)** *2013-2015*
  Viqarunnisa Noon School and College GPA: 5.00/5.00

Research Interest

- Applied Mathematics
- Data Science
- Mathematical Biology
- Bioinformatics

Publications

- **Nuzhat Nuari Khan**, Shahin Ara Begum, Raheedun Afeef, and Md Kamrujjaman. "Vaccine efficacy of COVID-19 in Bangladesh: Does vaccination prevent the pandemic?." GANIT: Journal of Bangladesh Mathematical Society 43, no. 1 (2023): 045-062.
- **Nuzhat Nuari Khan Rivu**, Md Kamrujjaman, and Asif Iqbal. "HIV/AIDS suppression in North America: Intervention plans and cost-effectiveness of UNAIDS 90-90-90 and 95-95-95 targets." Communications in Nonlinear Science and Numerical Simulation 152, Part B (2026): 109246.
- **Nuzhat Nuari Khan Rivu**, Md Kamrujjaman, and Shohel Ahmed. "Comparative analysis of stochastic and predictable models in the HIV epidemic across genders." *International Journal of Biomathematics* (Online Ready, 2025).

Professional Experience

- **Graduate Teaching Assistant** *Aug, 2024-Present*
  University of Iowa
- **Graduate Research Assistant, University Grant Commission (UGC)** *Apr, 2023-May, 2024*
  University of Dhaka
- **Mathematics Instructor** *Jan, 2019-May, 2024*
  Udvash Academic and Admission Care

Honors and Awards

- **National Science and Technology (NST) Fellowship** *2023*
  - **MS Thesis**, Group:Physical Science
- **Sumitomo Corporation Scholarship** *2020-2024*
  - The Sumitomo Corporation Scholarship provides financial assistance to outstanding university students in Bangladesh. The scholarship covers tuition, fees, and living expenses for one year. Scholarships are renewable for up to three years, subject to satisfactory academic progress.
- **Women's Mathematics Olympiad** *2019-2022*
  - **Women's Mathematics Olympiad Committee (WMOC)** of Dhaka University cooperated by **A F Mujibur Rahman Foundation (AFMRF)** organizes Women's Mathematics Olympiad for undergraduate female students of Bangladesh every year on 8th of March to celebrate the **International Women's Day**.

List of achievements:

2022: **1st** position
2021: **2nd** position
2020: **8th** position
2019: **6th** position

- **EWU Interuniversity Math Olympiad** *2019*
  - **4th** position
- **Board Scholarship-Dhaka Board** *2017*
  - Achieved Talentpool Scholarship (Merit Position: **60th** in Dhaka Board)

Certifications

- **Junior Certificate, Spanish Language (A1)** 1st division
Institute of Modern Language, University of Dhaka *Apr, 2022-Mar, 2023*
- **What is Data Science?**
IBM, Coursera *Aug, 2023*
- **Data Science: Foundations Using R Specialization**
IBM, Coursera *Jul, 2021*
- **Python for Everybody Specialization**
IBM, Coursera *Nov, 2020*

Affiliations

- **Organizing Volunteer** *Mar, 2023*
International Day of Mathematics
- **Organizing Secretary (Communication and Media)** *Jan, 2021-Jan, 2023*
Dhaka University IT Society (DUITS)
- **Membership Coordinator** *Jul, 2015-Mar, 2017*
Viqarunnisa Noon Debating Club

Expertise

Programming Languages and Softwares

- MATLAB
- LATEX
- Mathematica
- Fortran
- Python
- MS Excel

Soft Skills

- Teamwork
- Project and Event Management
- Presentation
- Leadership

References

Dr. Colleen Mitchell
Associate Professor
Department of Mathematics
University of Iowa
Email: colleen-mitchell@uiowa.edu

Dr. Laurent Jay
Professor
Department of Mathematics
University of Iowa
Email: laurent-jay@uiowa.edu